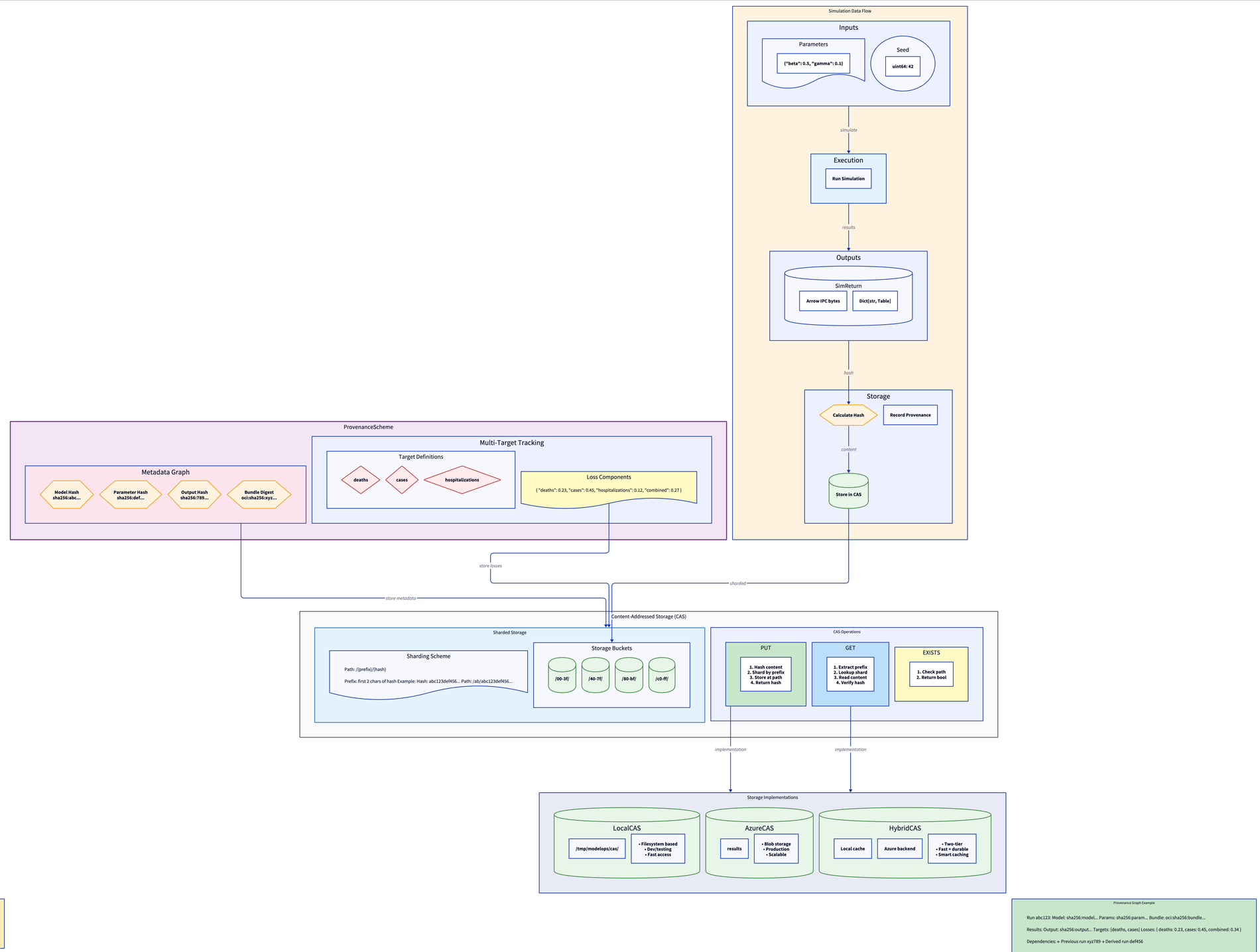
Simulation Data Flow
Inputs
Parameters
{"beta": 0.5, "gamma": 0.1}
Seed
uint64: 42
simulate
Execution
Run Simulation
results
Outputs
SimReturn
Arrow IPC bytes
Dict[str, Table]
hash
Storage
Calculate Hash
Record Provenance
content
Store in CAS
ProvenanceScheme
Multi-Target Tracking
Target Definitions
deaths
cases
hospitalizations
Loss Components
{ "deaths": 0.23, "cases": 0.45, "hospitalizations": 0.12, "combined": 0.27 }
Metadata Graph
Model Hash sha256:abc...
Parameter Hash sha256:def...
Output Hash sha256:789...
Bundle Digest oci:sha256:xyz...
store losses
sharded
store metadata
Content-Addressed Storage (CAS)
Sharded Storage
Sharding Scheme
Path: /{prefix}/{hash}
Prefix: first 2 chars of hash Example: Hash: abc123def456... Path: /ab/abc123def456...
Storage Buckets
/00-3f/
/40-7f/
/80-bf/
/c0-ff/
CAS Operations
PUT
1. Hash content
2. Shard by prefix
3. Store at path
4. Return hash
GET
1. Extract prefix
2. Lookup shard
3. Read content
4. Verify hash
EXISTS
1. Check path
2. Return bool
implementation
implementation
Storage Implementations
LocalCAS
/tmp/modelops/cas/
• Filesystem based
• Dev/testing
• Fast access
AzureCAS
results
• Blob storage
• Production
• Scalable
HybridCAS
Local cache
Azure backend
• Two-tier
• Fast + durable
• Smart caching
Benefits
✓ Complete reproducibility
✓ Deduplication (same content = same hash)
✓ Multi-target loss tracking
✓ Provenance graph navigation
✓ Immutable audit trail
Provenance Graph Example
Run abc123: Model: sha256:model... Params: sha256:param... Bundle: oci:sha256:bundle...
Results: Output: sha256:output... Targets: [deaths, cases] Losses: { deaths: 0.23, cases: 0.45, combined: 0.34 }
Dependencies: ← Previous run xyz789 → Derived run def456